

Kavi (Yomal Kavinda) Senewiratne

Robotics engineer and technical leader with 6+ years of experience in autonomous and humanoid robotic systems, spanning robot learning, planning, integration, and real-world deployment.

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RESEARCH INTERESTS

My research interest is in physical AI safety for humanoid robots, particularly in developing learned models that can predict the consequences of whole-body actions and support safe long-horizon planning in complex, multi-stage tasks.

EDUCATION

- **M.S. Robotics** 2020 – 2022
Oregon State University *Corvallis, OR, USA*
 - **Advisor:** Prof. Geoffrey A. Hollinger
 - **Thesis:** *Decentralized Multi-Robot Exploration with Sparse Communication through Deep Imitation Learning*
 - **Research focus:** Imitation learning, decentralized multi-robot exploration, path planning, and underwater robotics.
- **M.S. International Economics and Finance** 2016 – 2018
Technische Hochschule Nürnberg Georg Simon Ohm *Nuremberg, Germany*
 - **Advisor:** Prof. Dirk Honold
 - **Thesis:** *Fundraising Tools and Techniques for Valuating Startups*
- **B.Sc. Economics and Finance** 2012 – 2015
London School of Economics and Political Science *Colombo, Sri Lanka*

RESEARCH EXPERIENCE

- **Graduate Research Assistant** 04/2021 – 10/2022
Robotics Decision Making Laboratory, Oregon State University *Corvallis, OR, USA*
 - Developed an AlphaGo-inspired deep imitation learning approach for decentralized multi-robot exploration, letting robot teams keep coordinating when communication is sparse or drops out.
 - Built a two-dimensional underwater multi-robot simulator that generated synthetic training data at scale, removing the dependence on costly and scarce real-world underwater trials.
 - Developed a local path-planning pipeline for an autonomous underwater vehicle with the UW Applied Physics Laboratory and NAVFAC, reducing operator input for subsea turbine inspection.

INDUSTRY EXPERIENCE

- **Team Lead, Application Engineering** 07/2025 – Present
Senior Application Engineer 07/2024 – 07/2025
Apptронik *Austin, TX, USA*
 - Act as lead solutions architect for industrial humanoid deployments in the United States and Europe, designing the technical path from model evaluation to a working system on the customer floor.
 - Repeatedly led on-site teams of 5–10 engineers through multi-week industrial deployments, taking humanoid robots from on-site technical validation to productive work on live customer floors.
 - Defined the Applied AI Engineering roadmap for dataset analysis, model evaluation, and failure analysis, giving the team a repeatable path from trained model to deployed robot behavior.
- **Application Engineer** 10/2022 – 05/2024
Agility Robotics *Tangent, OR, USA*
 - Co-architected the Fleet Management System (FMS) that tasked and monitored robot fleets in live Amazon and Toyota warehouse operations, making continuous unattended production runs possible.
 - Implemented the company's first PLC-based workcell safety system and its FMS integration, verifying every state so robots halt correctly when something goes wrong; it became the reference template.

SELECTED ENTREPRENEURSHIP AND LEADERSHIP

- **Co-Founder and CEO** 01/2021 – 06/2022
pachaeearth Inc. *Corvallis, OR, USA*
 - Secured letters of intent from the Oregon Department of Forestry and the United States Forest Service, then built a wireless weather-monitoring and analytics platform for their prescribed burns.

- **Co-Founder and CEO** 12/2016 – 01/2021
Whammychat UG *Nuremberg, Germany*
 - Secured \$200,000 in pre-seed funding backed by six customer letters of intent, then led a six-person product and engineering team across software development, product direction, and hiring.

HONORS, AWARDS AND CERTIFICATION

- **Functional Safety of Machinery Engineer** 2025
TÜV Rheinland
 - Completed the qualifying course and examination in machinery functional safety, risk reduction, and safety-system architecture.
- **Start?Zuschuss!** 2019 – 2020
Bavarian Ministry of Economic Affairs, Regional Development and Energy
 - Received a grant totaling \$41,000 through a technology-oriented startup support program.
- **EXIST Gründerstipendium** 2018 – 2019
Federal Ministry for Economic Affairs and Energy and European Social Fund
 - First international student at Technische Hochschule Nürnberg to secure a \$140,000 grant supporting the development of a research-based and commercially viable technology product.

TECHNICAL SKILLS

- **Languages:** Python, C++, TypeScript
- **Machine learning:** PyTorch, scikit-learn, pandas, NumPy
- **Robotics and simulation:** ROS, Gazebo
- **Functional safety:** Safety PLCs, SICK Safety Designer
- **Tooling and infrastructure:** Linux, Git, Bazel, pytest, PostgreSQL